

Regional ICT Skills Championship 2025

IT Business Software Solutions

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1 Instructions

1.1 Introduction

This document describes the tasks to be performed for the Regional ICT Skills Championship 2025 in the category IT Business Software Solutions. We advise you to read this document carefully and to have an overview of the provided media like attached images and text. In the appendix you will find, besides other documents, the General Description which has been published before the competition and which should be known to you already. Please use it as a refresher as it contains important information for the competition, especially the rules and the tools available on the virtual machine in which the end product is tested for evaluation.

You may find yourself in the situation that there is not enough time to complete all remaining tasks. You can skip the implementation of certain tasks in favor for others. However, you are expected to deliver a smoothly running application at the end of the competition to facilitate the assessment of your work.

If you're not sure about how to interpret certain requirements, make an assumption and make sure to state your assumption in the readme. Your stated assumptions will be considered during the evaluation.

1.2 Rules

Please refer to the rules defined in the General Description.

1.3 Tools

Please refer to the tools listed in the General Description.

1.4 Evaluation

Please refer to the Evaluation chapter in the General Description.

1.5 Delivery / Handing in the results

Consult Chapter 4 Delivery for details on how to prepare and hand in the deliverables. **It is advised to reserve enough time for completing the competitor info, readme and preparing the deliverables.**

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2 Project Introduction

Many people go to the gym and perform exercises to stay fit. To help track workouts and progress, you are tasked with developing a gym tracking desktop application called AthliTrack.

2.1 Gym Tracking

Gym tracking involves documenting exercises performed during a training session. Each exercise typically consists of multiple **sets**, with each set containing a specific number of **repetitions (reps)**. Multiple sets together form a **workout**. To simplify tracking, workouts can be created using **templates**.

2.2 Application Structure

The application consists of multiple windows. Refer to the appendix for additional details.

2.3 Design

Wireframes in the appendix provide a visual reference for understanding the requirements. They are suggestions only, and your implementation does not need to match them exactly to earn points. Be aware that some aspects required for full points may not be fully represented in the wireframes.

2.4 Database

For this exercise, the application's data must persist, meaning it should remain available even after closing and reopening the application.

Ensure that the connection to your datastore or database works correctly within the VM, as all evaluations will take place there. If any additional setup is required, include clear instructions in the README.

2.5 General Requirements

The general requirements are valid for the whole application where not stated otherwise.

- All windows, except modals, are resizable.
 - The content is fitted well within the window frame.
 - The minimum window size is set to a reasonable size.
 - An appropriate overflow strategy is chosen to prevent cut content.
 - Errors in forms should be displayed in an intuitive way
 - Any undoable deletion requires explicit user confirmation. (The popular “are you sure that you want to delete this?” modal)
 - Modals for editing will not apply any changes until the save button is used. Cancelling or closing the modal will revert any made changes, after the user confirms the loss of unsaved changes.
 - A sophisticated exception handling prevents unhandled exceptions, especially program crashes.
- NOTE: *If an undocumented exception results in a crash and therefore prevents the test of a feature, no points are rewarded for that feature!*

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3 Implementation

3.1 Database & Import

Create a database and derive the structure from the given task description and wireframes.

Afterwards import the exercises given in the JSON file *exercises.json* with the following structure:

- name: string
- description: string
- type: string

3.2 Main Window

The first window shows the created workouts:

- Using the menu-strip the user can navigate to following pages:
 - o *Exercises Window*
 - o *Templates Window*
- All workouts are shown with following information:
 - o Workout name
 - o Start datetime
 - o End datetime
 - o Duration in hours and minutes
- It should be possible to delete workouts from this window
- Navigate to the *View Workout Window* when pressing the *View Workout* button
- Navigate to the *Add / Edit Workout Window* upon following actions:
 - o When pressing the *Add Workout* button
 - o When selecting a workout and pressing the *Edit Workout* button

3.3 Exercises Window

This window shows all exercises:

- All exercises are shown with following details:
 - o Name
 - o Description
 - o Type
- When pressing the *My Progress* button for an exercise, the user should be navigated to the *Progress Window* for chosen exercise
- It should be possible to delete exercises from this window
- It should be possible to add and edit exercises from this window
 - o All fields are mandatory

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3.4 Templates Window

This window shows all templates:

- All templates are shown with following details:
 - Name
- It should be possible to delete templates from this window
- Navigate to the *Add / Edit Template Window* upon following actions:
 - When pressing the *Add* button
 - When selecting a template and pressing the *Edit* button

3.5 Add / Edit Template Window

This window allows the user to add or edit a template:

- The page title is either:
 - *Add template* when adding a template
 - *Edit template "[template name]"* when editing a template
- The name field is a mandatory unique string
- It should be possible to add exercises to the workout with the following mandatory details:
 - Exercise (must be selected from the available exercises)
 - Amount of sets
 - Number of reps for each set of the exercise
- A Template must contain at least one exercise including at least one set and rep per set
- The order of exercises must be preserved
- It should be possible to edit the exercise's set and rep numbers
- It should be possible to remove an exercise from the workout
- Exercises must be unique within a template
- Cancel the process when pressing the *Cancel* button
- Save changes when pressing the *Save* button

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3.6 Add / Edit Workout Window

This window allows the user to add or edit a workout:

- When adding a workout, please consider the following:
 - o Upon starting, open a modal to select a template for pre-filling exercises
 - Sets and reps are loaded from the template
 - Weight (kg) must be entered manually
 - o The user can also start without a template, adding exercises manually
 - o Display an appropriate message if no templates are available
- The page title is either:
 - o *Add workout* when adding a workout
 - o *Edit workout "[workout name]"* when editing a workout
- The name field is a mandatory unique string
- Start and end are mandatory datetimes
 - o End must be after start
 - o The time of the workout must not overlap with other workouts
- Users should be able to add exercises to a workout
 - o Exercises must be selected from those available in the database
 - o For each exercise, the following details must be configurable:
 - Sets (automatically assigned when adding weight/reps)
 - Weight (kg)
 - Reps
 - o Exercise details are displayed in a table format:
 - Sets (row) numbers increment automatically, starting at 1, and cannot be manually adjusted.
 - Weight and reps can be edited inline by clicking their respective cells.
 - New sets are added by filling in a new row.
 - Sets can be deleted by swiping left.
 - The order of exercises and their details must be maintained.
- A workout must contain at least one exercise
- The exercises inside a workout must be unique
- Cancel the process when pressing the *Cancel* button
- Save changes when pressing the *Save* button

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3.7 View Workout Window

This window allows the user to view a workout:

- The title of the page must be *Workout "[Workout Name]"*
- The exercises must be displayed with the following details:
 - Exercise name
 - Sets
 - Weight
 - Reps
- When pressing the *My Progress* button for an exercise, the user should be navigated to the *Progress Window* for the chosen exercise
- Following statistics of the workout should be shown:
 - Total volume - Calculated by summing the weight lifted per set:
 - The weight lifted per set is defined by multiplying the weight with reps
 - Total amount of sets
 - Total amount of reps

3.8 Progress Window

This window shows the progress of an exercise:

- The title of the page must be *Progress of [Exercise Name]*
- List all executions of the selected exercise, displaying:
 - Execution date
 - Total volume
- If the exercise has been executed multiple times in different workouts of the same day, treat each session as a separate record
- Calculate the next expected total volume using the following approach:
 1. Determine the increase between each occurrence in chronological order
 2. Compute the average increase
 3. Add the average increase to the most recent total volume to estimate the next expected value
- When no records are available, display an appropriate message
- Display a line chart visualizing the trend of the progress including the next expected value
- For the data per se and calculation show the data of all time
- For the line chart show the data of the last 3 months only

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4 Delivery

With the new ruleset, it is hard to predict what project structures the competitors will have. Therefore, a simple delivery archive structure has been defined. Please make sure to comply with the structure and do not forget to include all relevant files. **The compliance with these formal requirements will be considered for evaluation!**

Your delivery file should be a .zip archive containing following structure:

- **source** (folder, where all your source code and material is contained)
- **executable** (folder, where your executable file including all necessary resources for running the application are found)
- **database** (folder, where all your database relevant files such as dumps are located)
- **competitor.json** (json-file, where you fill in your personal info. Template can be found in the projects archive)
- **readme.md** (textfile, where you list detailed instructions on how to run the application and may give some hints for the evaluation team to consider)

The filename of your archive should be “skill09-2025-**firstname-lastname**.zip”.

In the project archive you will find a template folder that already has the expected structure and files you can use for creating your deliverables.

You are free to add any files or folders which are needed for evaluation or installing/executing the program. Just make sure to put all mandatory files in their mandatory places.

Make sure to have your archive ready for delivery by the end of the competition and be ready to deliver your product in the way the supervisor instructs you to.

5 Appendix

You also find all individual documents within the “documents” folder of the projects files archive.

- Application structure (2 pages)
- Wireframes (5 pages)
- General description (3 pages)
- JSON file containing exercises